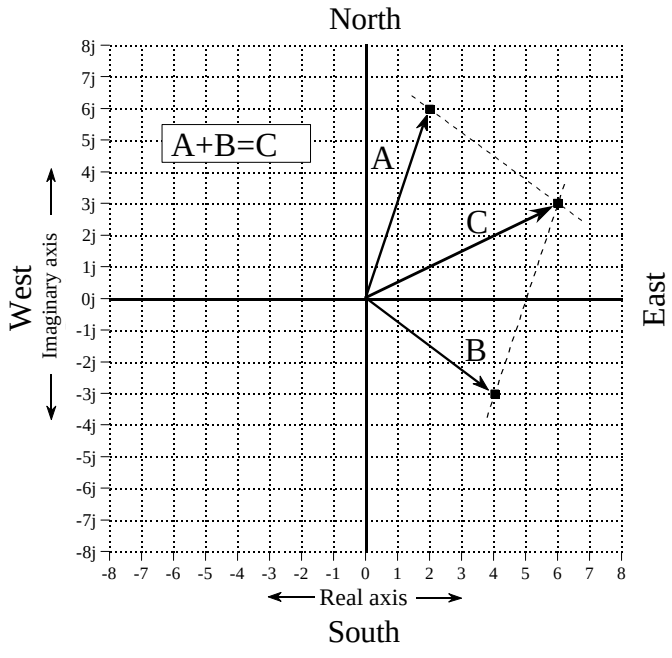

FIGURE 30-3
Adding vectors with complex numbers. The vectors A & B represent forces measured with respect to north/south and east/west. The east/west dimension is replaced by the real part of the complex number, while the north/south dimension is replaced by the imaginary part. This substitution allows complex mathematics to be used with an entirely *real* problem.

physical problem? For instance, we used Eq. 30-1 to add the force vectors in the sailboat problem. How do we know that the addition of complex numbers provides the same result as the addition of force vectors? In most cases, we know that complex mathematics can be used for a particular application because *someone else* said it does. Some brilliant and well respected mathematician or engineer worked out the details and published the results. The point to remember is that we cannot substitute just *any* problem into a complex form and expect the answer to make sense. We must stick to applications that have been shown to be applicable to complex analysis.

Let's look at an example where complex number substitution *does not* work. Imagine that you buy apples for \$5 a box, and oranges for \$10 a box. You represent this by the complex number: $5 + 10j$. During a particular week, you buy 6 boxes of apples and 2 boxes of oranges, which you represent by the complex number: $6 + 2j$. The total price you must pay for the goods is equal to number of items multiplied by the price of each item, that is, $(5 + 10j)(6 + 2j) = 10 + 70j$. In other words, the complex math indicates you must pay a total of \$10 for the apples and \$70 for the oranges. The problem is, the answer is completely wrong! The rules of complex mathematics *do not* follow the rules of this particular physical problem.

Complex Representation of Sinusoids

Complex numbers find a niche in electronics and signal processing because they are a compact way to represent and manipulate the most useful of all waveforms: sine and cosine waves. The conventional way to represent a sinusoid is: $M\cos(\omega t + \phi)$ or $A\cos(\omega t) + B\sin(\omega t)$, in polar and rectangular